

Prathmesh Jain

+91 8657376595 | prathmeshjain42@gmail.com | [LinkedIn](#) | [GitHub](#) | [LeetCode](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Full-Stack Engineer with experience building scalable web applications and backend systems using React, Node.js, and FastAPI. Skilled in designing APIs, distributed services, and data-driven applications with strong fundamentals.

EXPERIENCE

Tata Consultancy Services (TCS)

Oct 2024 – Present

Full Stack Engineer (Research & Innovation, CTO Team, BFSI Americas)

Mumbai, IN

- Built asynchronous workflow systems using **FastAPI, React, and LLM integrations** to automate compliance screening, document processing, and payment repair operations across multiple internal platforms.
- Developed concurrent backend pipelines for Google Search API calls, web crawling, document extraction, and verification tasks, enabling long-running processes to execute independently while improving overall system responsiveness.
- Implemented agent-driven workflows using LLM-based orchestration for adverse media analysis, credit memo generation, and payment repair flows involving multi-step validation and tool-based reasoning.
- Designed real-time streaming interfaces using Server-Sent Events to stream workflow progress, intermediate outputs, and asynchronous task updates directly to frontend applications.
- Built high-throughput extraction and verification pipelines using Playwright-based crawling, multithreaded document processing, and LLM-assisted analysis across large unstructured datasets and financial reports.
- Developed persisted and resumable workflow flows with rollback handling, validation checkpoints, and human review stages to support long-running financial spread generation and transaction repair operations.
- Implemented reliability patterns including retries, fallbacks, circuit breakers, exponential backoff, and graceful degradation across AI-based systems to improve fault tolerance and user experience during partial failures.
- Worked across frontend and backend systems using React, FastAPI, PostgreSQL, Zustand/Redux, RBAC, authentication flows, file uploads, and Docker-based deployments while contributing as a primary developer across multiple projects.

PERSONAL PROJECTS

Patronix – E-commerce Platform | *React.js, Node.js, Express.js, MongoDB*

[Live](#) | [GitHub](#)

- Built and deployed a full-stack e-commerce platform with live hosting, secure authentication, product catalog management, cart, payment gateway integration and order tracking flows.
- Optimized application performance and global state handling using Redux Toolkit.
- Designed modular, maintainable RESTful APIs with token-based authentication and clean architecture.

ExpenseGauge – Expense Tracking App | *React Native (Expo), Node.js, MongoDB*

[Live](#) | [GitHub](#)

- Designed and shipped a cross-platform, offline-first expense management application with real users and a downloadable Android apk.
- Implemented optimistic UI updates and background sync queues to ensure seamless usage during network outages.
- Implemented secure authentication using short-lived access tokens and rotating refresh tokens for seamless session continuity.
- Developed admin dashboards for user management, transaction audits, and balance adjustments.
- Built role-based backend services supporting expense analytics with monthly trends and category-wise visualizations.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java

Backend: Node.js, Express.js, FastAPI, Flask

Frontend: React.js, Next.js, React Native, Redux Toolkit, Zustand, TailwindCSS

Database: MongoDB, PostgreSQL

Tools: Docker, Git, Postman, Playwright

EDUCATION

Vidyalankar Institute of Technology

Mumbai, IN

Bachelors in Computer Engineering

May 2024

GPA: 9.2 / 10